UM HACKATHON 2026 · TEAM BOGOSORT

reclaim

AI-native expense reimbursement. HR effort mathematically bounded to **exceptions only.**

LangGraph | GLM-5.1 | GLM-4.6-Vision | FastAPI | Next.js 16 | PostgreSQL

01 / THE PROBLEM

100%

CLAIMS REVIEWED MANUALLY

5-10 Days

TURNAROUND TIME

0

AUTOMATED FRAUD SIGNALS

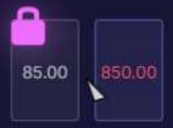
THE RECLAIM ENGINE

PASSED

AI verified. Zero violations. Fast-tracked.

ATTENTION

Policy breach flagged. HR provided with exact clause.



FRAUD TRAP

Immutable audit trail. Every edit logged and mathematically verified.

"Efficiency by Exception. AI classifies every claim before HR opens a single case."

03 / SYSTEM ARCHITECTURE (TWO-PHASE)

PHASE 1: POLICY INITIALIZATION

HR Admin → Upload Policy → AI Extractor → Policy DB → Active Policy

PHASE 2: CORE CLAIM EXECUTION

Employee Upload

OCR Processing

- Extract Data from Receipt
- User Verification
- Send to Agent

Compliance Agent

- Receive RAW JSON
- Compare with DB
- Call External Tools

Final Verdict

- HR Dashboard
- Flags & Logs

Status: Approve / Reject

04 / AGENTIC CORE

Policy Agent

Unstructured PDF → Structured JSON

Document Agent

Parallel OCR Processing

Compliance Agent

ReAct Evaluation Loop

05 / EXECUTION

LIVE DEMO

Reclaim running end-to-end in production.

EMPLOYEE PORTAL	AI PIPELINE	HR PORTAL
Upload → Verify → Submit	OCR → Compliance → Verdict	Triage → Review → Decide

06 / UI — TWO ROLE-ISOLATED PORTALS

Employee Portal

📤 Claim Submission Wizard

Category select → drag-and-drop upload → real-time OCR status per file

🔍 Side-by-Side Verification (Fraud Trap)

AI form (left) + original document (right). Any edit writes to audit trail immediately.

📊 Status Dashboard + Claim History

PENDING · APPROVED · PARTIAL · REJECTED with per-receipt AI reasoning inline

HR Portal

📋 Policy Studio

Upload PDF → AI extracts categories + conditions → HR confirms → status: ACTIVE

🎯 Triage Dashboard

Bucket A (Passed) separated from Bucket B (Attention). HR opens only what needs a human.

✅ Decision Engine

Full audit trail · Force Approve · Approve Adjusted · Confirm Rejection · HR Note

07 / THE X-RAY — FRAUD TRAP MECHANICS

Human Override Detection

RM 200.00

⚠️ Delta Detected: +900%

AI Flag Confidence

Image Integrity Analysis

```
> analyzing_pixels...
> ERROR: Artifact mismatch at X,Y
Possible photoshop overlay.
```

REJECTED

Shipped MVP

Parallel OCR
LIVE

Fraud Trap
LIVE

Policy Studio
LIVE

ReAct Engine
LIVE

Future Enhancement

- Direct Banking Integration
- Third-Party Messaging Bots
- Visual Drag-and-Drop

Workflow Builders

09 / THE TEAM

TEAM BOGOSORT

Darrell | Filbert | Christian | Michael | Chingiz

Thank you.

Built in 10 days for UMHackathon 2026.
Every claim: AI-analyzed, policy-grounded, permanently auditable.

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